

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

Name: Muhammad Usman

Roll No: 23BSCYS006

Score: _____ Signature of the Lab Tutor: _____ Date: _____

VLAN Routing

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT	SCORE
OBJECTIVE NO:				

Learning Objectives:

As a result of this lab section, you should achieve the following tasks:

- Configure VLAN interfaces□
- Establish VLAN routing on a single switch□
- Perform VLAN routing over an Ethernet Trunk link□
- Perform dynamic routing between VLAN interfaces using OSPF□

Keywords: VLAN Routing, OSPF, Switch Virtual Interface

Duration: 03 hours

Introduction

This lab introduces students to configuring VLANs and enabling communication between them using routing techniques on Huawei's eNSP network simulator. The exercises are designed to provide hands- on experience with both basic VLAN configuration and advanced routing mechanisms, preparing students to design and troubleshoot scalable enterprise networks.

Students will start by creating VLAN interfaces, then proceed to configure routing within a single switch, extend connectivity using Ethernet trunk links, and finally implement dynamic routing (OSPF) to automate route exchange between VLANs.

1. What is VLAN?

A VLAN (Virtual Local Area Network) is a logical grouping of devices within a switch network, regardless of their physical location.

Devices in the same VLAN behave as if they are on the same LAN

Devices in different VLANs are placed in separate broadcast domains

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

A VLAN is used to **segment a network** for:

- Better security
- Reduced broadcast traffic
- Easier network management

2. What is VLAN Routing?

VLAN Routing, also called Inter-VLAN Routing, is the technique used to enable communication between devices that belong to different VLANs.

Since VLANs logically separate a network into different broadcast domains, devices in different VLANs cannot communicate by default. A Layer 3 device (Router or Layer-3 Switch) is required to route traffic between them.

Why VLAN Routing is Needed

- VLANs isolate traffic for security, performance, and management
- But users in different VLANs often need to communicate (e.g., HR Server VLAN, Student VLAN)
- VLAN routing provides controlled and secure communication

How VLAN Routing Works

1. A device in VLAN 10 sends data to a device in VLAN 20
2. The source device forwards traffic to its default gateway
3. The Layer 3 device:
4. Receives the frame
5. Removes Layer-2 header
6. Routes the packet using IP routing
7. Packet is forwarded to the destination VLAN
8. Destination device receives the data

3. VLAN Interface (Switch Virtual Interface – SVI)

A VLAN Interface, commonly referred to as a Switch Virtual Interface (SVI), is a logical interface created on a network switch to represent a specific VLAN (Virtual Local Area Network) at Layer

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB

EXPERIMENT # 06

3. Unlike a physical port, an SVI is not tied to any single switch port it exists virtually and provides IP-based connectivity to an entire VLAN.

- An SVI allows a switch to perform routing between VLANs by assigning an IP address to the VLAN.□
- It's not a physical port on the switch.□
- For each VLAN that requires Layer 3 connectivity, one SVI is created, but only one SVI can be active per VLAN.□

How does it work?

- **Assign IP address to SVI:** VLAN 10 → IP 192.168.10.1/24
- **Hosts in VLAN 10 use this IP as their gateway:** Any device in VLAN 10 configured with IPs like 192.168.10.x will use 192.168.10.1 (the SVI) as their default gateway.
- **Switch performs routing between SVIs:** If VLAN 10 and VLAN 20 both have SVIs with different IP subnets, the switch can route between them without needing an external router.

Pros

- **Simplifies Inter-VLAN Routing**
- **Centralized VLAN Management**
- **Enables Layer 3 Switching**

4. Dynamic Routing with OSPF

OSPF (Open Shortest Path First) is a link-state dynamic routing protocol used to automatically exchange routing information between routers in an IP network. It dynamically calculates the shortest and most efficient path for data packets using the Dijkstra Shortest Path First (SPF) algorithm. OSPF is widely used in enterprise networks due to its scalability, fast convergence, and support for hierarchical network design.

How does OSPF work?

5. Routers send Hello packets on OSPF-enabled interfaces to discover and form adjacencies with neighbors.

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB

EXPERIMENT # 06

6. Each router floods LSAs containing details about its links (interfaces, cost, state).
7. LSAs are stored in the Link-State Database (LSDB).
8. Using the LSDB, each router runs the Dijkstra SPF algorithm to calculate the shortest path to all networks.
9. The calculated shortest paths are installed in the routing table for packet forwarding.

Pros

- **Open standard** multi-vendor support.
- **Efficient and scalable** for large networks.
- **Fast convergence** after topology changes.
- **Supports complex hierarchies** with areas.
- **No routing loops** due to SPF calculation.

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

Topology

Scenario:

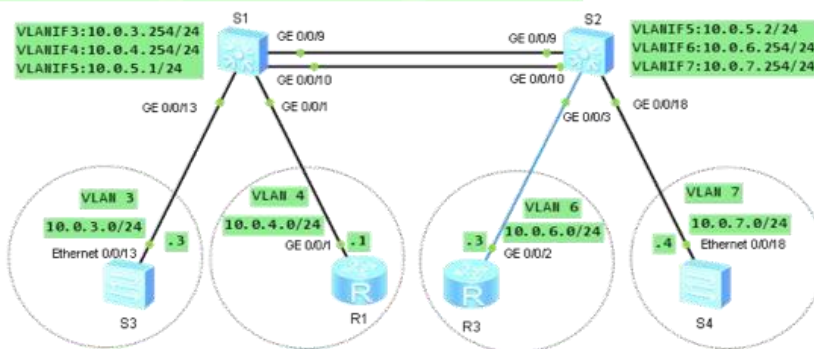
The introduction of layer three switches into the enterprise network opened up opportunities for streamlining the current VLAN routing configuration. The network administrator has been given the task to implement VLAN routing using only the layer three switches to support communication between the VLANs in the network as displayed in the topology. VLANs should be capable of inter VLAN communication. Additionally S1 and S2 are expected to communicate over a Layer 3 for which routing protocol support is required.

Learning Objectives:

As part of this lab section, you are expected to achieve the following tasks:

1. Configure VLAN interfaces.
2. Establish VLAN routing on a single switch.
3. Perform VLAN routing over an Ethernet Trunk link.
4. Perform dynamic routing between VLAN interfaces using OSPF.

```
[S1]dis vl
[S1]dis ip ro
```



```
<Huawei>system-view
[Huawei]sysname R1
[R1]int g0/0/1
[R1-GigabitEthernet0/0/1]ip add 10.0.4.1 24
[R1-GigabitEthernet0/0/1]q
[R1]ip ro 0.0.0.0 0.0.0.0 10.0.4.254
```

```
<Huawei>system-view
[Huawei]sysname S3
[S3]int e0/0/13
[S3-Ethernet0/0/13]shutdown
[S3-Ethernet0/0/13]undo shutdown
[S3-Ethernet0/0/13]q
[S3]int Vlanif 1
[S3-Vlanif1]ip add 10.0.3.3 24
[S3-Vlanif1]q
[S3]ip ro 0.0.0.0 0.0.0.0 10.0.3.254
```

```
[S1]ospf
[S1-ospf-1]area 0
[S1-ospf-1-area-0.0.0.0]network 10.0.0.0 0.255.255.255
-----
[S2]ospf
[S2-ospf-1]area 0
[S2-ospf-1-area-0.0.0.0]network 10.0.0.0 0.255.255.255
```

```
<Huawei>system-view
[Huawei]sysname R3
[R3]int g0/0/2
[R3-GigabitEthernet0/0/2]ip add 10.0.6.3 24
[R3-GigabitEthernet0/0/2]q
[R3]ip route-static 0.0.0.0 0.0.0.0 10.0.6.254
```

```
<Huawei>system-view
[Huawei]sysname S4
[S4]int e0/0/18
[S4-Ethernet0/0/18]shutdown
[S4-Ethernet0/0/18]undo shutdown
[S4-Ethernet0/0/18]q
[S4]int Vlanif 1
[S4-Vlanif1]ip add 10.0.7.4 24
[S4-Vlanif1]q
[S4]ip ro 0.0.0.0 0.0.0.0 10.0.7.254
```

```
Test connectivity between S3 and R1.
<R1>ping 10.0.3.3
Test connectivity between R3 and R1.
<R1>ping 10.0.6.3
The connectivity between R1 and R3 fails.
[R1]tracert 10.0.6.3
```

```
<Huawei>system-view
[Huawei]sysname S1
[S1]vlan batch 3 to 7
[S1]int Eth-Trunk 1
[S1-Eth-Trunk1]mode lacp-static
[S1-Eth-Trunk1]port link-type trunk
[S1-Eth-Trunk1]port trunk allow-pass vlan all
[S1-Eth-Trunk1]port trunk pvid vlan 5
[S1-Eth-Trunk1]q
[S1]int g0/0/9
[S1-GigabitEthernet0/0/9]eth-trunk 1
[S1-GigabitEthernet0/0/9]int g0/0/10
[S1-GigabitEthernet0/0/10]eth-trunk 1
[S1-GigabitEthernet0/0/10]q
[S1]int g0/0/1
[S1-GigabitEthernet0/0/1]port link-type access
[S1-GigabitEthernet0/0/1]port default vlan 4
[S1-GigabitEthernet0/0/1]q
[S1]int g0/0/13
[S1-GigabitEthernet0/0/13]port link-type access
[S1-GigabitEthernet0/0/13]port default vlan 3
[S1-GigabitEthernet0/0/13]q
[S1]int Vlanif 3
[S1-Vlanif3]ip add 10.0.3.254 24
[S1-Vlanif3]int v1 4
[S1-Vlanif4]ip add 10.0.4.254 24
[S1-Vlanif4]int v1 5
[S1-Vlanif5]ip add 10.0.5.1 24
```

```
<Huawei>system-view
[Huawei]sysname S2
[S2]vl b 3 to 7
[S2]int e 1
[S2-Eth-Trunk1]mo la
[S2-Eth-Trunk1]p l t
[S2-Eth-Trunk1]p t a v a
[S2-Eth-Trunk1]p t p v 5
[S2-Eth-Trunk1]q
[S2]int g0/0/9
[S2-GigabitEthernet0/0/9]e 1
[S2-GigabitEthernet0/0/9]int g0/0/10
[S2-GigabitEthernet0/0/10]e 1
[S2]int g0/0/3
[S2-GigabitEthernet0/0/3]p l a
[S2-GigabitEthernet0/0/3]p d v 6
[S2-GigabitEthernet0/0/3]q
[S2]int g0/0/18
[S2-GigabitEthernet0/0/18]p l a
[S2-GigabitEthernet0/0/18]p d v 7
[S2-GigabitEthernet0/0/18]q
[S2]int Vlanif 5
[S2-Vlanif5]ip add 10.0.5.2 24
[S2-Vlanif5]int v1 6
[S2-Vlanif6]ip add 10.0.6.254 24
[S2-Vlanif6]int v1 7
[S2-Vlanif7]ip add 10.0.7.254 24
[S2-Vlanif7]q
```

S1 does not have a route to the network segment 10.0.6.0 because the network segment is not directly connected to S1

Scenario

The introduction of layer three switches into the enterprise network opened up opportunities for streamlining the current VLAN routing configuration. The network administrator has been given the task to implement VLAN routing using only the layer three switches to support communication between the VLANs in the network as displayed in the topology. VLANs should be capable of inter VLAN communication. Additionally S1 and S2 are expected to communicate over a Layer 3 for which routing protocol support is required.

Tasks

Step 1 Preparing the environment

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

```
<Huawei> system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei] sysname R1
```

```
[R1] interface GigabitEthernet 0/0/1
```

```
[R1-GigabitEthernet0/0/1] ip address 10.0.4.1 24
```

```
<Huawei> system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei] sysname R3
```

```
[R3] interface GigabitEthernet 0/0/2
```

```
[R3-GigabitEthernet0/0/2] ip address 10.0.6.3 24
```

```
<Huawei> system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei] sysname S1
```

```
[S1] interface Eth-Trunk 1
```

```
[S1-Eth-Trunk1] mode lacp-static
```

```
[S1-Eth-Trunk1] port link-type trunk
```

```
[S1-Eth-Trunk1] port trunk allow-pass vlan all
```

```
[S1-Eth-Trunk1] quit
```

```
[S1] interface GigabitEthernet 0/0/9
```

```
[S1-GigabitEthernet0/0/9] eth-trunk 1
```

```
[S1-GigabitEthernet0/0/9] interface GigabitEthernet 0/0/10
```

```
[S1-GigabitEthernet0/0/10] eth-trunk 1
```

```
<Huawei> system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei] sysname S2
```

```
[S2] interface Eth-Trunk 1
```

```
[S2-Eth-Trunk1] mode lacp-static
```

```
[S2-Eth-Trunk1] port link-type trunk
```

```
[S2-Eth-Trunk1] port trunk allow-pass vlan all
```

```
[S2-Eth-Trunk1] quit
```

```
[S2] interface GigabitEthernet 0/0/9 [S2GigabitEthernet0/0/9]
```

```
eth-trunk 1
```

```
[S2-GigabitEthernet0/0/9] interface GigabitEthernet 0/0/10
```

```
[S2-GigabitEthernet0/0/10] eth-trunk 1
```

```
<Huawei> system-view
```

Enter system view, return user view with Ctrl+Z.

```
[Huawei] sysname S3
```

**DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06**

[S3]

```
<Huawei> system-view
Enter system view, return user view with Ctrl+Z.
[Huawei] sysname S4
[S4]
```

Step 2 Configure VLAN 3 through to VLAN 7 for S1 and S2

[S1] **vlan batch 3 to 7**

Info: This operation may take a few seconds. Please wait for a moment...done.

[S2] **vlan batch 3 to 7**

Info: This operation may take a few seconds. Please wait for a moment...done.

Verify that the VLANs have been created.

[S1] **display vlan**

The total number of vlans is : 6
...output omitted...
VID Type Ports

```
1 common UT:GE0/0/1(U) GE0/0/2(D) GE0/0/3(U) GE0/0/4(U)
    GE0/0/5(U) GE0/0/6(D) GE0/0/7(D) GE0/0/8(D)
    GE0/0/11(D) GE0/0/12(D) GE0/0/13(D) GE0/0/14(D)
    GE0/0/15(D) GE0/0/16(D) GE0/0/17(D) GE0/0/18(D)
    GE0/0/19(D) GE0/0/20(D) GE0/0/21(U) GE0/0/22(U)
    GE0/0/23(U) GE0/0/24(D) Eth-Trunk1(U)
```

3 common TG:Eth-Trunk1(U)

4 common TG:Eth-Trunk1(U)

5 common TG:Eth-Trunk1(U)

6 common TG:Eth-Trunk1(U)

7 common TG:Eth-Trunk1(U)

...output omitted...

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

```
[S2] display vlan
```

```
The total number of vlans is : 6
```

```
...output omitted...
```

```
VID Type Ports
```

```
1 common UT:GE0/0/1(U) GE0/0/2(D) GE0/0/3(U) GE0/0/4(U) GE0/0/5(U)
```

```
GE0/0/6(D) GE0/0/7(D) GE0/0/8(D)
```

```
GE0/0/11(U) GE0/0/12(U) GE0/0/13(U) GE0/0/14(D) GE0/0/15(D) GE0/0/16(D) GE0/0/17(D)
```

```
GE0/0/18(D) GE0/0/19(D) GE0/0/20(D) GE0/0/21(U) GE0/0/22(U)
```

```
GE0/0/23(U) GE0/0/24(D) Eth-Trunk1(U)
```

```
3 common TG:Eth-Trunk1(U) 4
```

```
common TG:Eth-Trunk1(U)
```

```
5 common TG:Eth-Trunk1(U) 6
```

```
common TG:Eth-Trunk1(U)
```

```
7 common TG:Eth-Trunk1(U)
```

```
...output omitted...
```

Step 3 Set the Eth-Trunk link between S1 and S2 with PVID 5

Add interfaces Gigabit Ethernet 0/0/1 and 0/0/13 of S1 to VLAN 4 and VLAN 3 respectively.

For S2, add interfaces Gigabit Ethernet 0/0/3 and G0/0/18 to VLAN 6 and VLAN 7 respectively.

```
[S1] interface Eth-Trunk 1
```

```
[S1-Eth-Trunk1] port trunk pvid vlan 5
```

```
[S1-Eth-Trunk1] quit
```

```
[S1] interface GigabitEthernet 0/0/1
```

```
[S1-GigabitEthernet0/0/1] port link-type access
```

```
[S1-GigabitEthernet0/0/1] port default vlan 4
```

```
[S1-GigabitEthernet0/0/1] quit
```

```
[S1] interface GigabitEthernet 0/0/13
```

```
[S1-GigabitEthernet0/0/13] port link-type access
```

```
[S1-GigabitEthernet0/0/13] port default vlan 3
```

```
[S2] interface Eth-Trunk 1
```

```
[S2-Eth-Trunk1] port trunk pvid vlan 5
```


DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

```
[S2-Eth-Trunk1] quit
[S2] interface GigabitEthernet 0/0/3
[S2-GigabitEthernet0/0/3] port link-type access
[S2-GigabitEthernet0/0/3] port default vlan 6
[S2-GigabitEthernet0/0/3] quit
[S2] interface GigabitEthernet 0/0/18
[S2-GigabitEthernet0/0/18] port link-type access
[S2-GigabitEthernet0/0/18] port default vlan 7
```

Run the **display vlan** command to view the configuration

```
[S1] display vlan
The total number of vlans is : 6
...output omitted...
VID Type Ports

-----
1 common UT:GE0/0/2(D) GE0/0/3(U) GE0/0/4(U) GE0/0/5(U)
    GE0/0/6(D) GE0/0/7(D) GE0/0/8(D) GE0/0/11(D)
    GE0/0/12(D) GE0/0/14(D) GE0/0/15(D) GE0/0/16(D)
    GE0/0/17(D) GE0/0/18(D) GE0/0/19(D) GE0/0/20(D)
    GE0/0/21(U) GE0/0/22(U) GE0/0/23(U) GE0/0/24(D) TG:Eth-
    Trunk1(U)
3 common UT:GE0/0/13(U)
    TG:Eth-Trunk1(U)
4 common UT:GE0/0/1(U)
    TG:Eth-Trunk1(U)
5 common TG:Eth-Trunk1(U)
6 common TG:Eth-Trunk1(U)
7 common TG:Eth-Trunk1(U)

...output omitted...
```

Step 4 Configure gateway addresses for VLANs on S1 and S2

Configure IP addresses for VLANIF 3, VLANIF 4, and VLANIF 5 on S1, and for VLANIF 5, VLANIF 6, and VLANIF 7 on S2.

```
[S1] interface Vlanif 3
[S1-Vlanif3] ip address 10.0.3.254 24
```

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

```
[S1-Vlanif3] interface Vlanif 4
[S1-Vlanif4] ip address 10.0.4.254 24
[S1-Vlanif4] interface Vlanif 5
[S1-Vlanif5] ip address 10.0.5.1 24
[S2] interface Vlanif 5
[S2-Vlanif5] ip address 10.0.5.2 24
[S2-Vlanif5] interface Vlanif 6
[S2-Vlanif6] ip address 10.0.6.254 24 [S2-
Vlanif6] interface Vlanif 7
[S2-Vlanif7] ip address 10.0.7.254 24
```

Step 5 IP addressing and default routes for R1, R3, S3, and S4

```
[R1] ip route-static 0.0.0.0 0.0.0.0 10.0.4.254
[S3] interface Vlanif 1
[S3-Vlanif1] ip address 10.0.3.3 24
[S3-Vlanif1] quit
[S3] ip route-static 0.0.0.0 0.0.0.0 10.0.3.254
[R3] interface GigabitEthernet 0/0/2
[R3-GigabitEthernet0/0/2] ip address 10.0.6.3 24
[R3-GigabitEthernet0/0/2] quit
[R3] ip route-static 0.0.0.0 0.0.0.0 10.0.6.254
[S4] interface Vlanif 1
[S4-Vlanif1] ip address 10.0.7.4 24
[S4-Vlanif1] quit
[S4] ip route-static 0.0.0.0 0.0.0.0 10.0.7.254
```

Step 6 Test connectivity between VLAN 3 and VLAN 4

Test the connectivity between S3 and R1

```
<R1> ping 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=254 time=37 ms
  Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=253 time=2 ms
  Reply from 10.0.3.3: bytes=56 Sequence=3 ttl=253 time=10 ms
```

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB

EXPERIMENT # 06

```
Reply from 10.0.3.3: bytes=56 Sequence=4 ttl=253 time=3 ms
Reply from 10.0.3.3: bytes=56 Sequence=5 ttl=253 time=2 ms
```

```
--- 10.0.3.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
```

```
round-trip min/avg/max = 2/10/37 ms
```

Step 7 Enable OSPF on S1 and S2

The Open Shortest Path First (**OSPF**) protocol, developed by the Internet Engineering Task Force (IETF), is a link-state Interior Gateway Protocol (IGP). At present, OSPF Version 2, defined in RFC 2328, is intended for IPv4, and OSPF Version 3, defined in RFC 2740, is intended for IPv6.

```
[S1] ospf
[S1-ospf-1] area 0
[S1-ospf-1-area-0.0.0.0] network 10.0.0.0 0.255.255.255
[S2] ospf
[S2-ospf-1] area 0
[S2-ospf-1-area-0.0.0.0] network 10.0.0.0 0.255.255.255
```

After the configuration, wait until **S1** and **S2** exchange OSPF routes and complete the link state database, then view the resulting routing table of **S1**.

```
[S1] display ip routing-table
Route Flags: R - relay, D - download to fib
```

Routing Tables: Public

Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.3.0/24	Direct	0	0	D	10.0.3.254	Vlanif3
10.0.3.254/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.4.0/24	Direct	0	0	D	10.0.4.254	Vlanif4
10.0.4.254/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.5.0/24	Direct	0	0	D	10.0.5.1	Vlanif5
10.0.5.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
10.0.6.0/24	OSPF	10	2	D	10.0.5.2	Vlanif5

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

10.0.7.0/24	OSPF	10	2	D	10.0.5.2	Vlanif5
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0

S1 has learned two routes using **OSPF**.

Again test the connectivity between **R1** and **R3**.

```
<R1> ping 10.0.6.3
```

```
PING 10.0.6.3: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.6.3: bytes=56 Sequence=1 ttl=253 time=11 ms
```

```
Reply from 10.0.6.3: bytes=56 Sequence=2 ttl=253 time=1 ms
```

```
Reply from 10.0.6.3: bytes=56 Sequence=3 ttl=253 time=10 ms
```

```
Reply from 10.0.6.3: bytes=56 Sequence=4 ttl=253 time=1 ms
```

```
Reply from 10.0.6.3: bytes=56 Sequence=5 ttl=253 time=1 ms
```

```
--- 10.0.6.3 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

```
round-trip min/avg/max = 1/4/11 ms
```

```
<R1> ping 10.0.7.4
```

```
PING 10.0.7.4: 56 data bytes, press CTRL_C to break
```

```
Reply from 10.0.7.4: bytes=56 Sequence=1 ttl=253 time=30 ms
```

```
Reply from 10.0.7.4: bytes=56 Sequence=2 ttl=252 time=2 ms
```

```
Reply from 10.0.7.4: bytes=56 Sequence=3 ttl=252 time=3 ms
```

```
Reply from 10.0.7.4: bytes=56 Sequence=4 ttl=252 time=2 ms
```

```
Reply from 10.0.7.4: bytes=56 Sequence=5 ttl=252 time=2 ms
```

```
--- 10.0.7.4 ping statistics ---
```

```
5 packet(s) transmitted
```

```
5 packet(s) received
```

```
0.00% packet loss
```

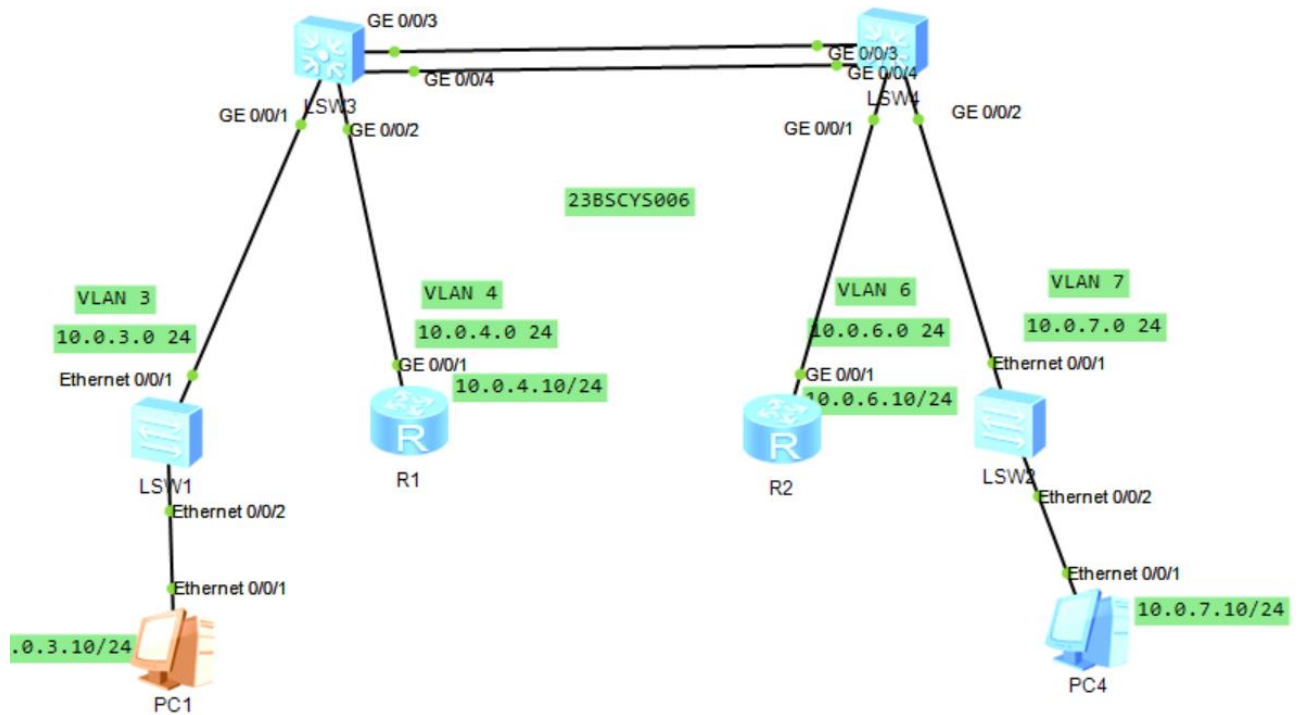
```
round-trip min/avg/max = 2/7/30 ms
```

Tasks

Try different topology with different interface numbers.

Upload the **.topo** **.efz** **.cfg** files.

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06



DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB

EXPERIMENT # 06

ID	Type	Ports			
	common	UT:GE0/0/5 (D)	GE0/0/6 (D)	GE0/0/7 (D)	GE0/0/8 (D)
		GE0/0/9 (D)	GE0/0/10 (D)	GE0/0/11 (D)	GE0/0/12 (D)
		GE0/0/13 (D)	GE0/0/14 (D)	GE0/0/15 (D)	GE0/0/16 (D)
		GE0/0/17 (D)	GE0/0/18 (D)	GE0/0/19 (D)	GE0/0/20 (D)
		GE0/0/21 (D)	GE0/0/22 (D)	GE0/0/23 (D)	GE0/0/24 (D)
		TG:Eth-Trunk1 (U)			
	common	UT:GE0/0/1 (U)			
		TG:Eth-Trunk1 (U)			
	common	UT:GE0/0/2 (U)			
		TG:Eth-Trunk1 (U)			
	common	UT:Eth-Trunk1 (U)			
	common	TG:Eth-Trunk1 (U)			
	common	TG:Eth-Trunk1 (U)			
ID	Status	Property	MAC-LRN	Statistics	Description
	enable	default	enable	disable	VLAN 0001
	enable	default	enable	disable	VLAN 0003
	enable	default	enable	disable	VLAN 0004
	enable	default	enable	disable	VLAN 0005
	enable	default	enable	disable	VLAN 0006
	enable	default	enable	disable	VLAN 0007
SW-3-006]dis eth					
SW-3-006]dis eth-trunk 1					
eth-Trunk1's state information is:					

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

```

VID      Status  Property      MAC-LRN Statistics Description
-----
1        enable  default      enable  disable  VLAN 0001
3        enable  default      enable  disable  VLAN 0003
4        enable  default      enable  disable  VLAN 0004
5        enable  default      enable  disable  VLAN 0005
6        enable  default      enable  disable  VLAN 0006
7        enable  default      enable  disable  VLAN 0007

[SW-3-006]dis eth
[SW-3-006]dis eth-trunk 1
Eth-Trunk1's state information is:
Local:
LAG ID: 1                      WorkingMode: STATIC
Preempt Delay: Disabled        Hash arithmetic: According to SIP-XOR-DIP
System Priority: 32768          System ID: 4clf-cc23-7181
Least Active-linknumber: 1     Max Active-linknumber: 8
Operate status: up             Number Of Up Port In Trunk: 2
-----
ActorPortName      Status  PortType  PortPri  PortNo  PortKey  PortState  Weight
GigabitEthernet0/0/3  Selected  1GE       32768    4       305      10111100   1
GigabitEthernet0/0/4  Selected  1GE       32768    5       305      10111100   1

Partner:
-----
ActorPortName      SysPri  SystemID      PortPri  PortNo  PortKey  PortState
GigabitEthernet0/0/3  32768   4clf-cc0e-749b  32768    4       305      10111100
GigabitEthernet0/0/4  32768   4clf-cc0e-749b  32768    5       305      10111100

```

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
Network Protocol & Architecture LAB
EXPERIMENT # 06

```
PC1
Basic Config | Command | MCPacket | UdpPacket | Console
Request timeout!
Request timeout!
From 10.0.7.10: bytes=32 seq=5 ttl=126 time=188 ms

--- 10.0.7.10 ping statistics ---
 5 packet(s) transmitted
 1 packet(s) received
80.00% packet loss
round-trip min/avg/max = 0/188/188 ms

PC>ping 10.0.7.10

Ping 10.0.7.10: 32 data bytes, Press Ctrl_C to break
From 10.0.7.10: bytes=32 seq=1 ttl=126 time=125 ms
From 10.0.7.10: bytes=32 seq=2 ttl=126 time=109 ms
From 10.0.7.10: bytes=32 seq=3 ttl=126 time=125 ms
From 10.0.7.10: bytes=32 seq=4 ttl=126 time=125 ms
From 10.0.7.10: bytes=32 seq=5 ttl=126 time=125 ms

--- 10.0.7.10 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 109/121/125 ms

PC>|
```

```
[R2-006]ping 10.0.3.10
PING 10.0.3.10: 56 data bytes, press CTRL_C to break
  Reply from 10.0.3.10: bytes=56 Sequence=1 ttl=126 time=110 ms
  Reply from 10.0.3.10: bytes=56 Sequence=2 ttl=126 time=110 ms
  Reply from 10.0.3.10: bytes=56 Sequence=3 ttl=126 time=110 ms
  Reply from 10.0.3.10: bytes=56 Sequence=4 ttl=126 time=90 ms
  Reply from 10.0.3.10: bytes=56 Sequence=5 ttl=126 time=110 ms

--- 10.0.3.10 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
round-trip min/avg/max = 90/106/110 ms

[R2-006]
```